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Lab#04 Introduction to SSB Modulation & Demodulation

Objective

The objective of this lab is to understand the concept of SSB modulation, demodulation and analyze it by using filter and phase shift method.

In-Lab:

Task 1: Design the model of SSB modulation and demodulation using Phase shift method to extract USB or LSB.

Task 2: Design the model of SSB using quadrature phase shift method.

Introduction

Single sideband, SSB modulation is basically a derivative of amplitude modulation, AM. By removing some of the components of the ordinary AM signal it is possible to significantly improve its efficiency.

It is possible to see how an AM signal can be improved by looking at the spectrum of the signal. When a steady state carrier is modulated with an audio signal, for example a tone of 1 kHz, then two smaller signals are seen at frequencies 1 kHz above and below the main carrier.

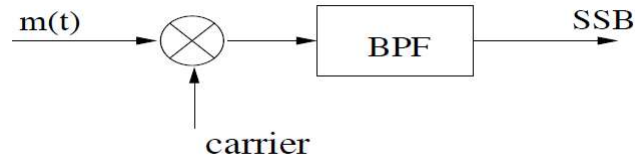
If the steady state tones are replaced with audio like that encountered with speech or music, these comprise many different frequencies and an audio spectrum with frequencies over a band of frequencies is seen. When modulated onto the carrier, these spectra are seen above and below the carrier.

It can be seen that if the top frequency that is modulated onto the carrier is 6 kHz, then the top spectra will extend to 6 kHz above and below the signal. In other words the bandwidth occupied by the AM signal is twice the maximum frequency of the signal that is used to modulate the carrier, i.e. it is twice the bandwidth of the audio signal to be carried.

Amplitude modulation is very inefficient from two points. The first is that it occupies twice the bandwidth of the maximum audio frequency, and the second is that it is inefficient in terms of the power used. The carrier is a steady state signal and in itself carries no information, only providing a reference for the demodulation process. Single sideband modulation improves the efficiency of the transmission by removing some unnecessary elements. In the first instance, the carrier is removed - it can be re-introduced in the receiver, and secondly one sideband is removed - both sidebands are mirror images of one another and they carry the same information. This leaves only one sideband - hence the name Single Side Band / SSB.

Bandpass filtering: One method of producing SSB signal is to remove one of the sidebands via filtering, leaving only either the upper sideband (USB), the sideband with the higher frequency, or less commonly the lower sideband (LSB), the sideband with the lower frequency. Most often, the carrier is reduced or removed entirely (suppressed), being referred to in full as single sideband suppressed carrier (SSBSC). Assuming both sidebands are symmetric, which is the case for a normal AM signal, no information is lost in the process. Since the final

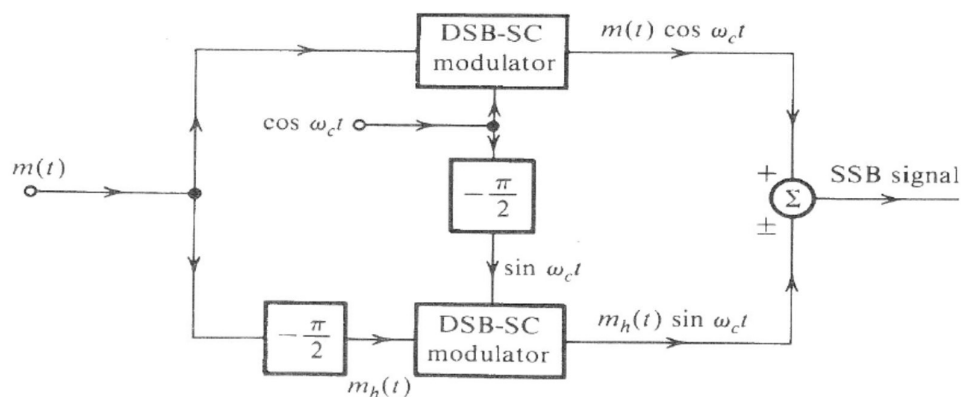
RF amplification is now concentrated in a single sideband, the effective power output is greater than in normal AM (the carrier and redundant sideband account for well over half of the power output of an AM transmitter). Though SSB uses substantially less bandwidth and power, it cannot be demodulated by a simple envelope detector like standard AM.



Hartley modulator (Phase shift method): An alternate method of generation known as a Hartley modulator, named after R. V. L. Hartley, uses phasing to suppress the unwanted sideband. To generate an SSB signal with this method, two versions of the original signal are generated, mutually 90° out of phase for any single frequency within the operating bandwidth. Each one of these signals then modulates carrier waves (of one frequency) that are also 90° out of phase with each other. By either adding or subtracting the resulting signals, a lower or upper sideband signal results. A benefit of this approach is to allow an analytical expression for SSB signals, which can be used to understand effects such as synchronous detection of SSB.

Shifting the baseband signal 90° out of phase cannot be done simply by delaying it, as it contains a large range of frequencies. In analog circuits, a wideband 90-degree phase-difference network^[7] is used. The method was popular in the days of vacuum tube radios, but later gained a bad reputation due to poorly adjusted commercial implementations. Modulation using this method is again gaining popularity in the homebrew and DSP fields. This method, utilizing the Hilbert transform to phase shift the baseband audio, can be done at low cost with digital circuitry.

$$\phi_{SSB}(t) = m(t) \cos(\omega_c t) \mp m_h(t) \sin(\omega_c t)$$



Demodulation of SSB

The demodulation of the SSB-SC signal is similar to the DSB-SC synchronous demodulation. The modulated SSB signal is multiplied by a signal with the same phase as the original carrier signal and then passed through a low pass filter.

In- Lab Task:

Task 1: Design the model of SSB modulation using Phase shift method to extract USB or LSB.

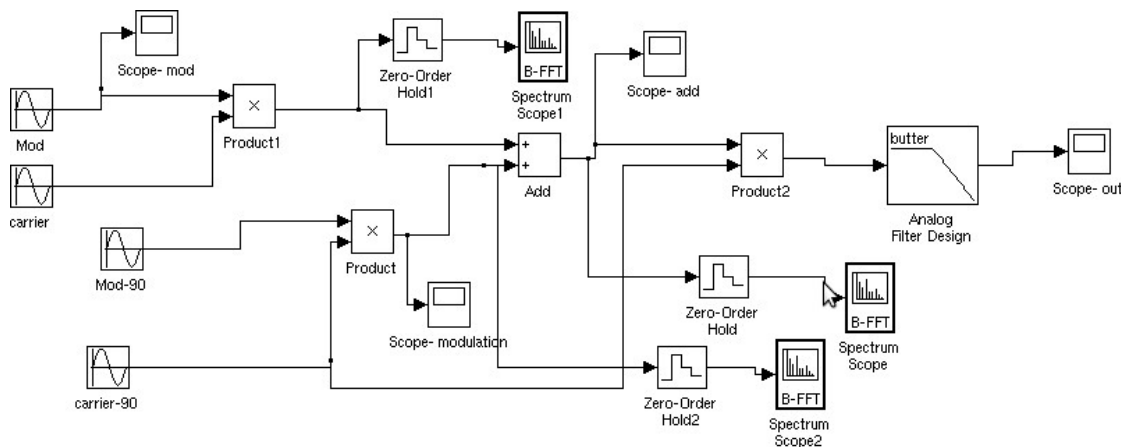


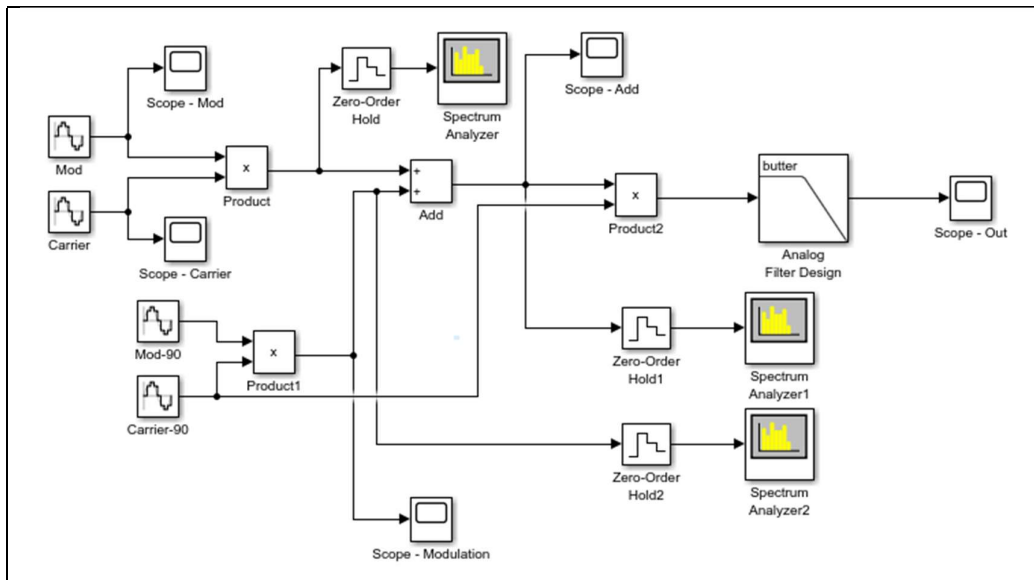
Figure 1

First set up the circuit as seen in Figure 1 on Simulink. The parameters are as follows:

- Modulating signal: frequency 1kHz, amplitude of 1V. (The signal fed into the lower multiplier is $f(t)$, which is $\pi/2$ phase shifted – labelled “Mod-90” in Figure 1.)
- Carrier signal: frequency 20kHz, amplitude of 1V. (The carrier signal fed into the lower multiplier is must also be $\pi/2$ phase shifted)
- Set the system sample rate need to satisfy the Nyquist criteria. The highest frequency component in the simulation is 41kHz, which occurs at the output of the demodulator. **What should be the sample rate to get correct frequencies at the output??** __At least 82 kHz as that is the double of the highest frequency component.
- Check the output of the two multipliers.
- Simulate from 0 to 10ms. This should show 10 cycles of the modulating waveform.

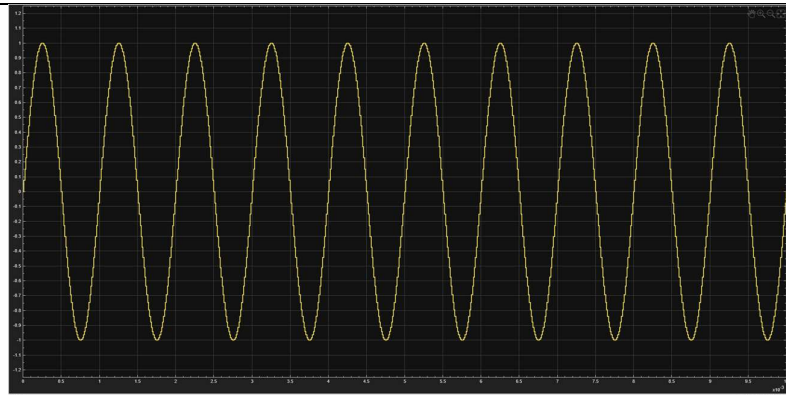
Write down the Amplitude and Frequency of Message Signal, Carrier, Modulated signal and demodulated signal. Also draw its waveform and Spectra of modulated signal.

Labeled Simulink model



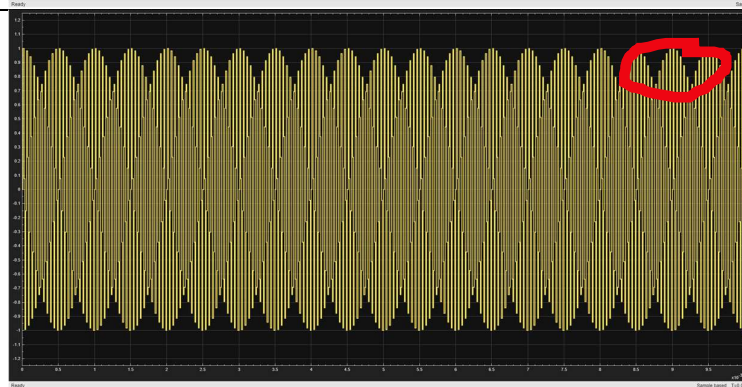
Message Signal

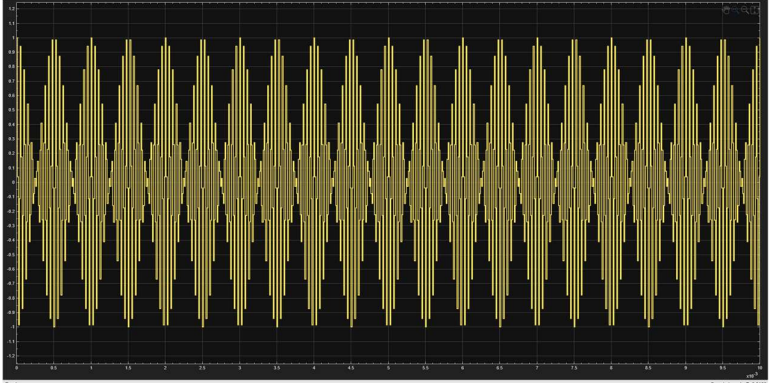
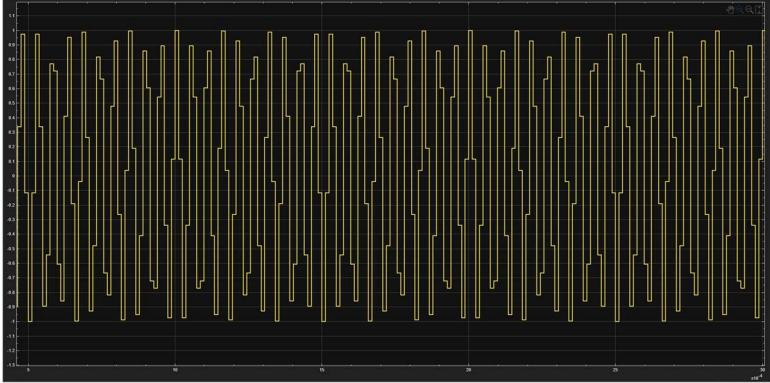
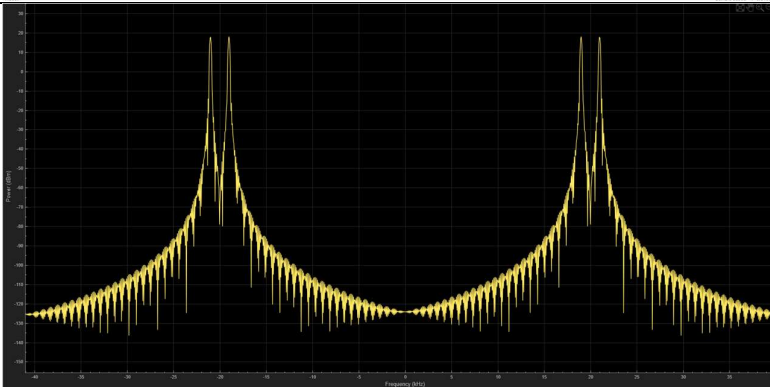
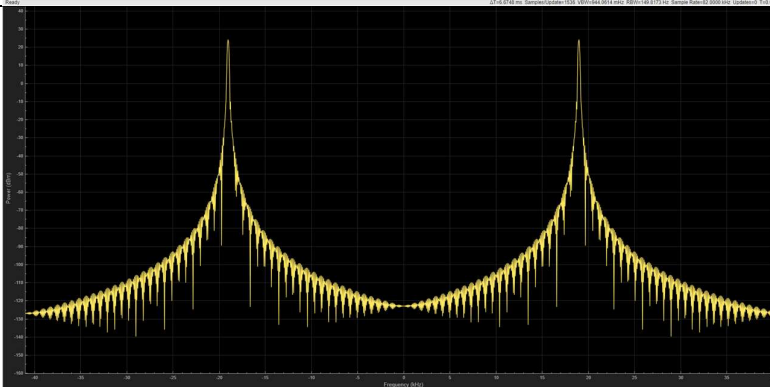
Amplitude = 1V
Frequency = 1kHz



Carrier Signal

Amplitude = 1V
Frequency = 20kHz



Modulated Signal Amplitude = 1V		
Output Signal after add block Amplitude = 1V		
Spectra of Signal after Product 1		
Spectra of SSB after Add block		



Task 2:

Quadrature Multiplexing

Generate two signals $f_1(t)$ and $f_2(t)$, one with frequency at 50Hz and the other with frequency 1kHz. Multiply one with a sine wave of frequency 20kHz and the other with a cosine of the same frequency as shown above in Figure 2.

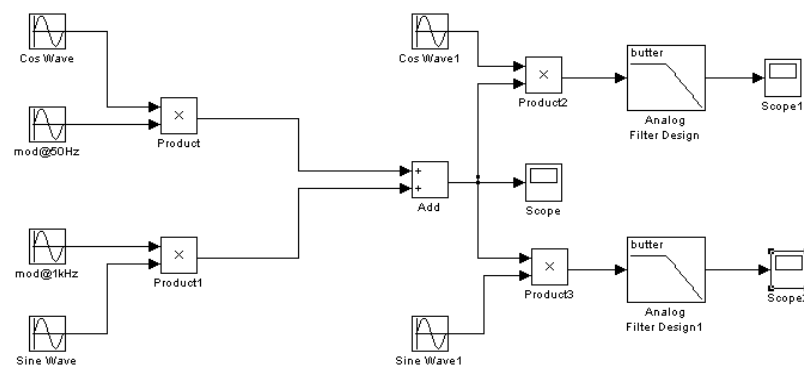


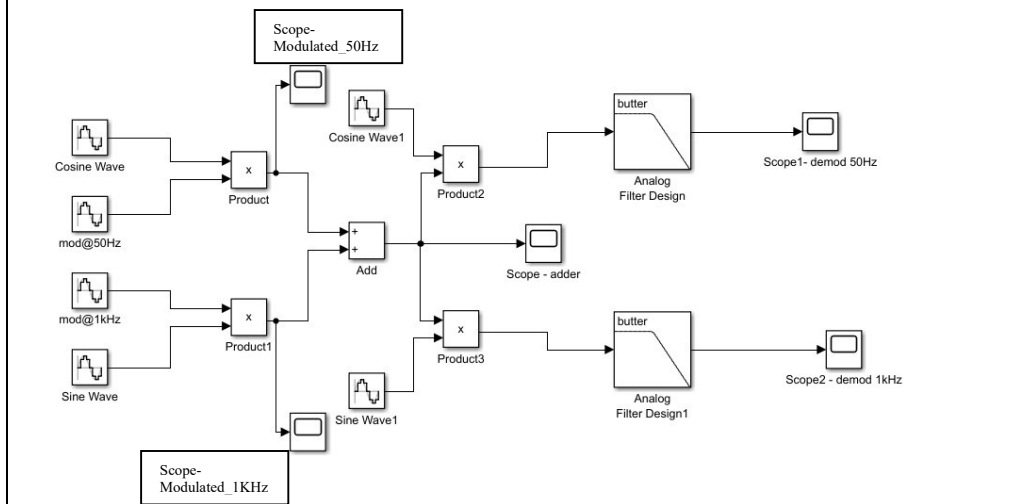
Figure 2

Sum the two modulated signals to obtain the multiplexed modulated carrier signal. Demodulate by multiplying by either a cosine wave followed by a low pass filter to recover $f_1(t)$, or by multiplying by a sine wave to recover $f_2(t)$ (create two additional signal generator). Check that the correctly demodulated signals appear at the two outputs. Note that these output signals have half the amplitude of the modulating waveforms $f_1(t)$ and $f_2(t)$ – this factor of a half is consistent with the mathematical analysis.

Now introduce a phase error into one (or both) of the demodulating carrier signals. What happens as you increase the phase error from 0 to 90 degrees?

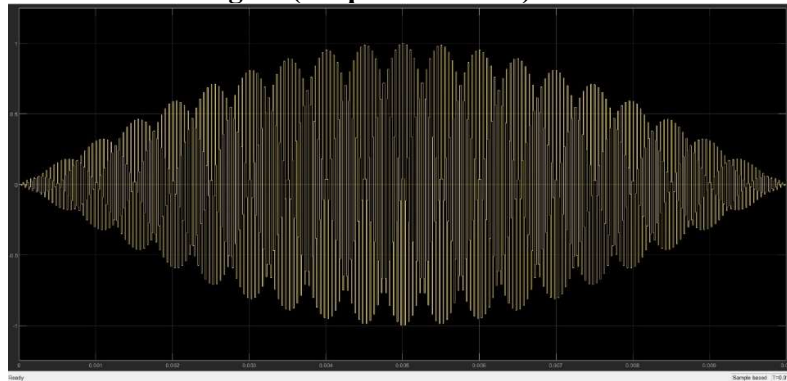
When we have no phase error, we get smooth signals $f_1(t)$ and $f_2(t)$. As we introduce and increase the phase error from 0° to 90° , signals $f_1(t)$ and $f_2(t)$ don't recovering correctly and we see many distortions in recovered signals.

Labeled Simulink model for task 2

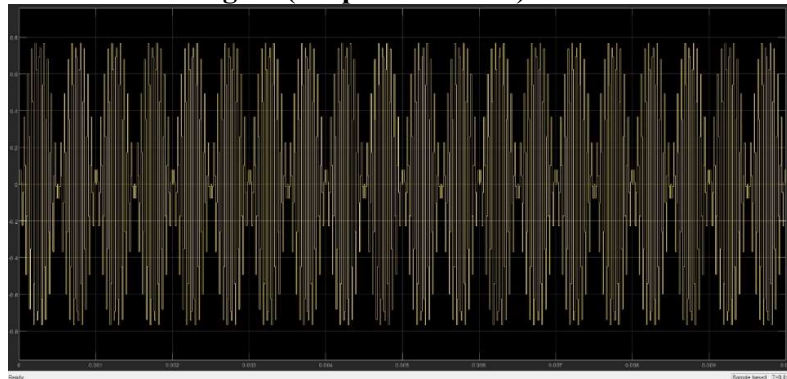


Modulated Signal

For 50 Hz mod signal (Amplitude = ~1V)



For 1kHz mod signal (Amplitude = ~1V)



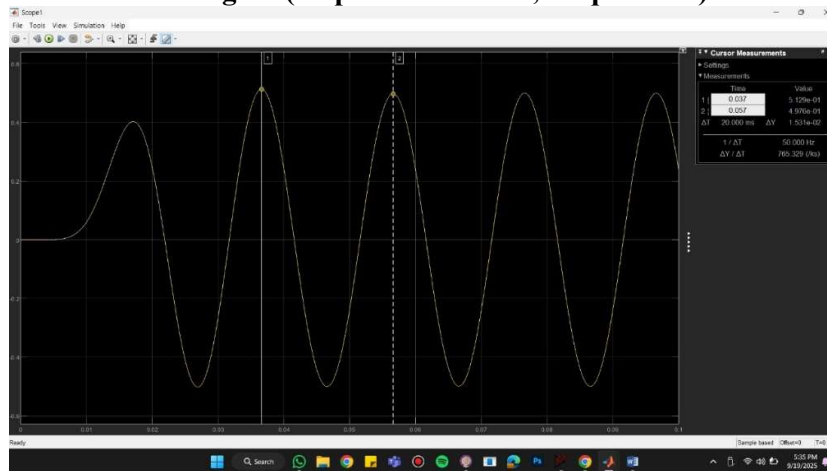
**Output Signal
after add block**

**Amplitude =
~1.5V**

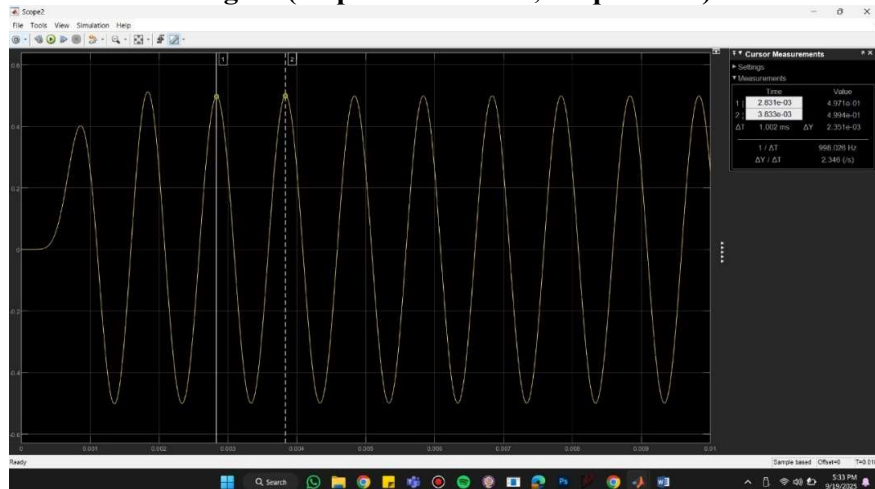


**Demodulated
Signal**

For 50Hz mod signal (amplitude = ~0.5V, freq = 50Hz)



For 1kHz mod signal (amplitude = ~0.5V, freq = 1kHz)



**Note: Attach the labelled Simulink model of both tasks.
(Both attached with their respective tasks above)**

Name & ID: _____

Date: _____

EE-322L Analog and Digital Communications Lab**Fall 2025****Habib University****Dhanani School of Science & Engineering****Lab#04 SSB Modulation and Demodulation****Lab #4 Marks distribution:**

		LR2=30	LR3=20	LR4=10	LR5=20	AR4=20
In-Lab Tasks	Task 1	/20	/10	-----	/10	/20
	Task 2	/10	/10	/10	/10	
Marks Obt.	98 /100					

Lab Evaluation Assessment Rubric

#	Assessment Elements	(0<Level 1<=4)	(4< Level 2<=6)	(6< Level 3<=8)	(8< Level 4<=10)
LR3	Troubleshooting	Unable to identify the fault/minimal effort shown	Able to identify the fault but unable to remove it.	Able to identify the fault but partially removes it.	Able to identify the fault and able to make necessary steps and actions to correct it.
LR4	Data Collection	Measurements are incomplete, inaccurate and imprecise. Observations are incomplete or not included. Symbols, units and significant figures are not included	Measurements are somewhat inaccurate and very imprecise. Observations are incomplete or recorded in a confusing way. Major errors using symbols, units and significant digits	Measurements are mostly accurate. Observations are generally complete. Minor errors using symbols, units and significant digits	Measurements are both accurate and precise. Observations are very thorough. Includes appropriate symbols, units and significant digits and completed in due time.
LR5	Results & Plots	Figures, graphs, tables contain errors or are poorly constructed, have missing titles, captions, units missing or incorrect, etc.	Most figures, graphs, tables OK, some still missing some important or required features	All figures, graphs, tables are correctly drawn, but some have minor problems or could still be improved	All figures, graphs, tables are correctly drawn and contain titles/captions.
AR4	*Report Submission	Late submission after 1 week.	Late submission after 4 days and within a week.	Late submission after the lab timing and after 2 days of the due date.	Timely submission of the report and in lab time.

***Report:** Report will not be accepted after due date.